P8131 Spring 2026 Homework #9 Solution

Due on April 26 at 11:59pm

1. Determine the survival and density functions for a continuous survival time variable with hazard function
$$h(x) = \frac{2x}{(1+x^2)}.$$
Hint: consider the derivative of $\log(1+x^2)$.
Solution:
$$H(t) = \int_0^t h(x)dx = \log(1+t^2),$$
$$S(t) = \exp(-H(t)) = \frac{1}{1+t^2},$$
$$F(t) = 1 - S(t) = \frac{t^2}{1+t^2},$$
$$f(t) = F^{'}(t) = \frac{2t}{(1+t^2)^2}.$$

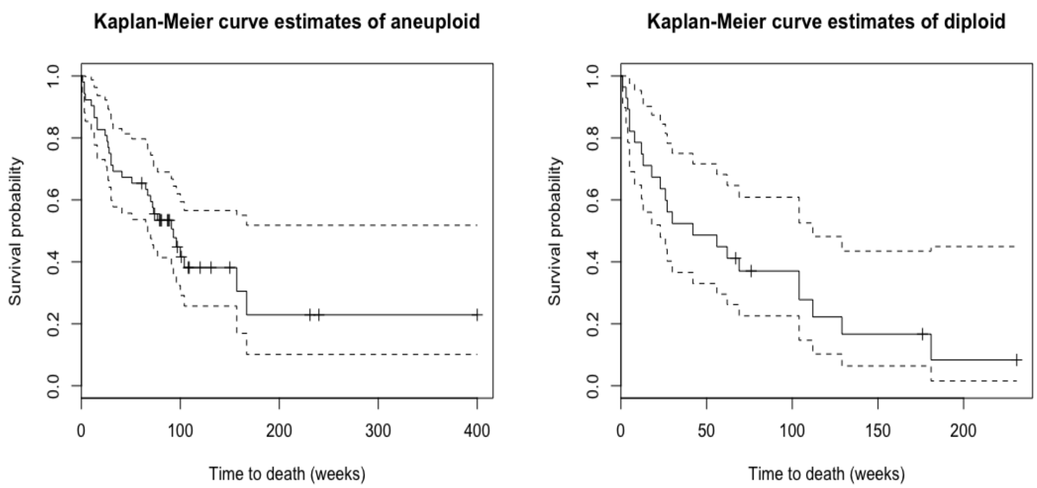
2. For the following data
$$1, 2, 2, 4+, 5+, 6, 7+, 8+, 9+, 10+,$$
where + denotes a right censored observation. Write out the data table and calculate the following by hand.
   (a) Find the Kaplan-Meier estimate of the survival function;
   (b) Find the Nelson-Aalen estimate of the cumulative hazard function;
   (c) Find the Fleming-Harrington estimate of the survival function.

Solution: K-M estimate: $\hat{S}(t)$; N-A estimate: $\hat{H}(t)$; F-H estimate: $\exp(-\hat{H}(t))$.

t_j	interval	n_j	d_j	$\frac{d_j}{n_j}$	$1 - \frac{d_j}{n_j}$	$\hat{S}(t)$	$\hat{H}(t)$	$\exp(-\hat{H}(t))$
1	[1, 2)	10	1	1/10	9/10	0.9	0.1	0.90
2	[2, 4)	9	2	2/9	7/9	0.7	0.322	0.72
4	[4, 5)	7	0	0	1	0.7	0.322	0.72
5	[5, 6)	6	0	0	1	0.7	0.322	0.72
6	[6, 7)	5	1	1/5	4/5	0.56	0.522	0.59
7	[7, 8)	4	0	0	1	0.56	0.522	0.59
8	[8, 9)	3	0	0	1	0.56	0.522	0.59
9	[9, 10)	2	0	0	1	0.56	0.522	0.59
10	[10, ∞)	1	0	0	1	0.56	0.522	0.59

3. Use the tongue data in the R package KMsurv. For each tumor type (aneuploidy and diploid), plot the Kaplan-Meier curve of survival function and its pointwise 95% confidence intervals (using the log transformation). What are the estimated 1-year survival rate and 95% CI?

   Solution:



For aneuploidy, 1-year survival estimate is 0.654, with 95% CI (0.537, 0.797).

For diploid, 1-year survival estimate is 0.486, with 95% CI (0.33, 0.716).

```
> # obtain survival rate at 1 yr=52 weeks, with CI
> summary(km_aneuploid,time=c(52))
Call: survfit(formula = survObj ~ type, data = aneuploid, conf.type = "log",
    type = "kaplan-meier")

 time n.risk n.event survival std.err lower 95% CI upper 95% CI
   52     34      18    0.654   0.066        0.537        0.797
> summary(km_diploid,time=c(52))
Call: survfit(formula = survObj ~ type, data = diploid, conf.type = "log",
    type = "kaplan-meier")

 time n.risk n.event survival std.err lower 95% CI upper 95% CI
   52     13      14    0.486  0.0961         0.33        0.716
```

Appendix

```
library(survival)
library(MASS)
library(KMsurv)

# Problem 2
time <- c(1, 2, 2, 4, 5, 6, 7, 8, 9, 10)
event <- c(1, 1, 1, 0, 0, 1, 0, 0, 0, 0)
survObj <- Surv(time, event==1)
km <- survfit(survObj~1, type="kaplan-meier")
summary(km)
```

```
plot(km)

# K-M, N-A, F-H
cbind(km$time,km$surv,cumsum(km$n.event/km$n.risk),
exp(-cumsum(km$n.event/km$n.risk)))


# Problem 3
data("tongue")
aneuploid$survObj <- with(aneuploid, Surv(time, delta==1))
aneuploid <- subset(tongue, type==1)
diploid <- subset(tongue, type==2)
aneuploid$survObj <- with(aneuploid, Surv(time, delta==1))
diploid $survObj <- with(diploid , Surv(time, delta==1))

km_aneuploid <- survfit(survObj~type, conf.type="log",
type="kaplan-meier", data=aneuploid )
summary(km_aneuploid )
plot(km_aneuploid , mark.time = T,
main="Kaplan-Meier curve estimates of aneuploid",
ylab = "Survival probability", xlab="Time to death (weeks)")

km_diploid <- survfit(survObj~type, conf.type="log",
type="kaplan-meier", data=diploid)
summary(km_diploid)
plot(km_diploid , mark.time = T,
main="Kaplan-Meier curve estimates of diploid",
ylab = "Survival probability", xlab="Time to death (weeks)")

# obtain survival rate at 1 yr=52 weeks, with CI
summary(km_aneuploid,time=c(52))
summary(km_diploid,time=c(52))
```